

The A-theory of time and the B-theory of time

PHIL UA-78 Metaphysics

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Two visions concerning the nature of time

The B-theory of time

a.k.a. the *Block Universe*

All times are *metaphysically on a par*

The facts about which times are *past*, *present*, and *future* are nothing over and above the facts about which times are before which.

Change is just a certain kind of pattern in the facts.

Time is like space

The A-theory of time

a.k.a. *Robust Passage*

The present time is *objectively metaphysically special*

There are *further* facts about which times are past, present, and future, over and above the facts about before and after.

The facts themselves undergo change.

Time is like modality

The key question

As I see it, the key point of disagreement between the two views concerns the following thesis:

Fact Eternalism

Every state of affairs that ever obtains, always obtains.

B-theorists accept Fact Eternalism: they think that whether a state of affairs obtains is an eternal matter. A-theorists reject it: they think that states of affairs can obtain temporarily.

- For the B-theorist, saying that a state of affairs 'sometimes obtains' or 'always obtains' is weird and misleading in the same way as saying that a number is 'sometimes prime' or 'always prime'. 'Obtains' is just not the sort of predicate which can interestingly be modified by 'sometimes' and 'always'.

The B-theory

How could anyone be a B-theorist?

The following is obviously true:

(1) NYC is sometimes sunny, but not always sunny.

Doesn't this immediately imply that there is a state of affairs—namely, *that NYC is sunny*—that obtains sometimes but not always?

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Doesn't this immediately imply that there is a state of affairs—namely, *that NYC is sunny*—that obtains sometimes but not always?

Not obviously!

Analogy with 'somewhere' and 'everywhere'

Compare:

(2) It is sunny somewhere but not everywhere.

Does it follow that there is a state of affairs—namely, *that it is sunny*—that obtains somewhere but not everywhere?

That would be weird! There isn't this spatially variable state of affairs *that it is sunny*, over and above all the different ordinary states of affairs of the form *it is sunny at such and such place*.

When I assert 'It is sunny', I express one of these ordinary states of affairs. Which one I express will depend on where I am (or on where I'm intending to talk about).

- ▶ Sentences and other bits of language like this are called *context-sensitive* or *indexical*.
- ▶ The word 'here' and sentences containing it are *obviously* context-sensitive: but the example of 'It is sunny' shows that some sentences that don't contain this word are context-sensitive in a very similar way.

Similarly, according to B-theorists, when I assert 'It is sunny in NYC', I express some state of affairs of the form *that it is sunny in NYC at t*. Which one I express will depend on when that utterance takes place (or, occasionally, on my intentions).

- ▶ This applies to typical sentences in the past, present, or future tense. B-theorists think that any sentence that could be used, literally, at one time to say something true and at another time to say something false, is context-sensitive.
- ▶ They'll also think 'now' is context-sensitive—though there is reason for A-theorists to agree with them about this. What's distinctive is that present-tense sentences without 'now' are taken to be context-sensitive in the same way.

How could anyone be a B-theorist?

OK, but how does that help us understand (1) and (2), or why they don't imply the existence of states of affairs whose obtaining varies across time or space?

For an even clearer analogy, consider:

- (3) Some farmer is such that he owns a donkey, but not every farmer is such that he owns a donkey.

This does not imply that there is a state of affairs—*that he owns a donkey*—such that some farmer is such that it obtains, but not every farmer is such that it obtains, just as 'Every farmer is such that he loves his mother' doesn't imply 'There is a person such that every farmer loves her'.

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The pronoun 'he' can be used *deictically* to refer to particular people (e.g., while pointing at them). But it also has a different kind of use that is available only when it occurs in the scope of a quantifier like 'Every farmer': it can be *bound* by the quantifier, like a variable in logic.

The hidden variable theory

One way of developing the semantics of tensed language in a B-theoretic way takes this analogy very seriously. We posit *hidden time-variables* associated with many ordinary predicates (e.g. tensed verbs, nouns, ...). These can be either *free* or *bound* by something higher up in the same sentence. When free, they function like deictic pronouns to refer to a time—the time of utterance.

Joe is happy.

Happy(Joe, t)

Whenever Sue is home, Joe is happy.

$\forall t(\mathbf{Home}(\mathbf{Sue}, t) \rightarrow \mathbf{Happy}(\mathbf{Joe}, t))$

Joe will be happy.

$\exists t(t > t' \wedge \mathbf{Happy}(\mathbf{Joe}, t))$

Joe has always been happy.

$\forall t(t < t' \rightarrow \mathbf{Happy}(\mathbf{Joe}, t))$

The A-theory

The A-theory and the analogy with modality

Recall that A-theorists affirm, while B-theorists deny, that (1) implies that there is a state of affairs that obtains sometimes but not always.

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The A-theory and the analogy with modality

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By contrast, it is pretty much uncontroversial that

(4) It is possibly the case, but not necessarily the case, that snow is white.

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Thus, for A-theorists, there is a fundamental parallelism between temporal modifiers (like ‘sometimes’ and ‘always’) and modal modifiers (like ‘possibly’ and ‘necessarily’).

The specialness of the actual world

On Lewis's view where possible worlds are parallel universes, there is nothing special about the actual world—'the actual world' is just what we call the world *we* happen to live in.

But on a regular theory of possible worlds, there is something very special about the actual world: it is the one and only world such that every state of affairs that obtains *at it*, obtains.

Of course the fact that it's special in this way is *contingent*. But that doesn't make it any less objective!

To say that there are centaurs in the real world, for example, is not to say that there are centaurs in some region of the universe in which we happen to have more interest than in others; it is simply to say that there are centaurs. Talk of the real world, in other words, is not a metaphorical fudging-up of talk in which our sentences have a special kind of prefix, but a fudging-up of talk in which the relevant sentences have no prefixes at all. "Really", "actually", "in fact", "in the real world" are strictly redundant expressions - that, and not any prejudice or provincialism, is their specialness. (Prior, 'The Notion of the Present', 321)

The A-theory and the specialness of the present

Similarly, according to A-theorists there is something very special about the present moment: it is the one and only *time* such that no state of affairs obtains *at it* but does not *obtain*.

Of course the fact that it's special in this way is *temporary*. But that doesn't make it any less objective!

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For B-theorists, by contrast, there is no more reason to think that the present moment is an objectively special time than to think that *here* is an objectively special place.

Prior on presentness

It is tempting to think of the present as a region of the universe in which certain things happen, such as the war in Vietnam, and the past and the future as other regions in which other things happen, such as the battle of Hastings and men going to Mars. But to this picture there is the same objection as to the picture of the "real world" as a box or region among other boxes or regions. It doesn't bring out what is so special about the present; and to be more specific, it doesn't bring out the way in which the present is real and the past and future are not. And I want to suggest that the reality of the present consists in what the reality of anything else consists in, namely the absence of a qualifying prefix. To say that Whitrow's lecture is past is to say that it has been the case that Whitrow is lecturing. To say that Scott's lecture is future is to say that it will be the case that Scott is lecturing. But to say that my lecture is present is just to say that I am lecturing — flat, no prefixes. (Prior, 'The Notion of the Present', 321–2)